

CS560 Framework

1. Framework Overview

This framework serves as a foundational base that you can modify for your assignments. It is built using OpenGL and GLSL for graphics, and utilizes the Assimp library to load FBX models.

2. Instructions

Camera Controls

You can control the camera using both the keyboard and the mouse:

a. Keyboard Controls:

- **Camera Movement:**
 - Use the numeric keypad:
 - 4: Move left
 - 8: Move up
 - 6: Move right
 - 2: Move down
- **Zoom:**
 - +: Zoom in
 - -: Zoom out
- **Rotation:**
 - 0: Rotate the camera
- **Initialization:**
 - 5: Reset the camera to its initial position

b. Mouse Controls:

- **Left Drag:** Rotate the camera around the Y-axis
- **Right Drag:** Move the camera position

3. Framework Structure

The framework consists of the following files:

```
framework
├── framework.h/cpp
├── fbo.h/cpp
├── gbuffer.h/cpp
├── interact.h/cpp
├── Mesh.h/cpp
├── scene.h/cpp
├── shader.h/cpp
├── shapes.h/cpp
├── textures.h/cpp
└── transform.h/cpp
```

4. Key Files to Work On

1. **framework.h/cpp**
 - Contains the main function of the application.
2. **scene.h/cpp**
 - Responsible for rendering everything visible on the screen, including the UI and models.
 - To create multiple scenes, you can derive a new class from the scene class.
3. **Mesh.h/cpp**
 - Contains the BoneData structure and the Mesh class, which handles loading mesh data from FBX files and rendering the models.

Additional Notes

- Sample FBX files can be found in the '**fbx**' folder.
- Textures for each model are located in the '**fbx/textures**' folder.
- Shader files are available in the '**shaders**' folder, where you can modify them as needed along with the source code.